Diego Gonzalez

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Education

University of Waterloo Sept. 2025 – April 2030
Bachelor of Mathematics, Honours, Data Science Major *Waterloo, ON*

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Probability, Linear Algebra, Multivariable Calculus, Algorithm Design & Data Abstraction

Technical Skills

Languages: Python, SQL, C++, C, Java, JavaScript, TypeScript
Data & ML: PySpark, PyTorch, NumPy, pandas, scikit-learn, Modal, OpenCV, Weights & Biases
Web: FastAPI, Flask, Next.js, Node.js, React, Vercel
Infra & Tools: Azure, Git/GitHub, Jupyter, Linux, PostgreSQL, Power BI, VS Code

Experience

Data Scientist Intern Jan 2027 – Apr 2027
Scotiabank *Toronto, ON*

- Incoming Winter 2027

Tech Lead & Consultant May 2026 – Present
Sun Strategy Group *Toronto, ON*

- Directed a **team of 3** across **6 client projects** for small and medium-sized businesses (**SMBs**) across **Ontario**.
- Owned the firm's tech workstream, leading recommendation calls with tech clients and designing custom websites that turned strategy into shipped assets.
- Designed and launched sunstrategygroup.ca, the firm's main platform showcasing five service lines and past projects.

Lead Instructor May 2026 – Sept. 2026
Code-It Hacks (Microsoft & Google Backed) *Toronto, ON*

- Designed and led project-based curriculums spanning **Python**, **ML**, **CV**, and **VR** development for **100+ students**.
- Translated advanced technical concepts into age-appropriate, hands-on lessons and gave students exposure to industry tools.
- Guided students through design and iterative problem-solving across multiple domains, strengthening their confidence.

Projects

Midas | *Python, JavaScript, Next.js, FastAPI, YOLOv11, OpenCV* Feb. 2026

- Developed a real-time AR repair assistant using **React** and **FastAPI** to integrate computer vision with an interactive UI.
- Reduced **OpenAI** and **YOLO API** calls by **99%** through a key-frame detection strategy, demonstrating production-minded reliability and inference optimization.
- Deployed AI service via **FastAPI** backend with scalability and low-latency responses in mind.

Duolingo for ASL | *Python, PyTorch, pandas, OpenCV, MediaPipe, SQLAlchemy* Jan. 2026

- Built an AI-powered ASL recognition app with a custom-trained **PyTorch** model on **70,000+** images, using feature engineering on **MediaPipe** hand landmarks to reach **90%+** accuracy on real-time inference.
- Designed a modular **ML** pipeline separating training, inference, and data processing, improving experiment iteration speed by **50%** and supporting rapid model evaluation and fine-tuning cycles.
- Implemented **SQLAlchemy** database to track user progress and manage data quality.

AI Chess Bot | *Python, PyTorch, Modal, Weights & Biases* Nov. 2025

- Built a chess engine powered by a **PyTorch** neural network achieving ∼**1600 ELO**, using **Weights & Biases** to track experiments, compare training runs, and guide iterative model evaluation and performance tuning.
- Improved move search efficiency by **50%** by implementing a minimax search algorithm with decision tree pruning.
- Reduced training time by **90%** by designing an end-to-end **ML** pipeline on **Modal** cloud infrastructure, enabling scalable batch training on large datasets.

Leadership & Extracurriculars

Education Assistant, UW Statistics Club Sept. 2026 – Present

- Shortened STAT230 and STAT231 review documents by **20%**, simplifying complex topics and removing redundancies.

Member, UW Data Science Club Sept. 2025 – Present

- **2nd** place in Jane Street x UWDSC Estimathon out of **350+** participants.

Head, North Toronto Computer Science Club Sept. 2023 – June 2025

- **Doubled** club membership by introducing beginner-friendly programming events and industry guest speakers.
- Prepared **50+** students for the Canadian Computing Competition, with **11** earning certificates of distinction.